CANONICAL ISOMORPHISM OMEGA_{G/S} WITH THE CONORMAL SHEAF OF THE IDENTITY SECTION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma morphisms-lemma-base-change-differentials (canonical tag 01V0). cited at proof lines 871-874. Differentials of the base change G x_S G over G.
- Lemma morphisms-lemma-differentials-relative-immersion-section (canonical tag 0474). cited at proof lines 894-896. Conormal sheaf equals e^* Omega.

Source reference. Original lemma 01V0: sources/stacks-chow/tags.txt, source lines 2429–2429. Cited at proof lines 871-874 to identify Omega_{G x_S G / G} with pr_0^* Omega; load-bearing.

Source reference. Original lemma 0474: sources/stacks-chow/tags.txt, source lines 5495–5495. Cited at proof lines 894-896 to get C_{S/G} = e^* Omega_{G/S}; load-bearing.

Lemma 0.2. *Let (G, m, e, i) be a group scheme over the scheme S. Denote $f : G \to S$ the structure morphism. Then there exist canonical isomorphisms*

$$\Omega_{G/S} \cong f^*\mathcal{C}_{S/G} \cong f^*e^*\Omega_{G/S}$$

where $\mathcal{C}_{S/G}$ denotes the conormal sheaf of the immersion e. In particular, if S is the spectrum of a field, then $\Omega_{G/S}$ is a free $\mathcal{O}_G$-module.

Proof. By Morphisms, Lemma 01V0 we have

$$\Omega_{G\times_S G/G} = \mathrm{pr}_0^*\Omega_{G/S}$$

where on the left hand side we view $G \times_S G$ as a scheme over G using pr_1. Let $\tau : G \times_S G \to G \times_S G$ be the "shearing map" given by $(g, h) \mapsto (m(g, h), h)$ on points. This map is an automorphism of $G \times_S G$ viewed as a scheme over G via the projection pr_1. Combining these two remarks we obtain an isomorphism

$$\tau^*\mathrm{pr}_0^*\Omega_{G/S} \to \mathrm{pr}_0^*\Omega_{G/S}$$

Since $\mathrm{pr}_0 \circ \tau = m$ this can be rewritten as an isomorphism

$$m^*\Omega_{G/S} \to \mathrm{pr}_0^*\Omega_{G/S}$$

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Pulling back this isomorphism by $(e \circ f, \mathrm{id}_G) : G \to G \times_S G$ and using that $m \circ (e \circ f, \mathrm{id}_G) = \mathrm{id}_G$ and $\mathrm{pr}_0 \circ (e \circ f, \mathrm{id}_G) = e \circ f$ we obtain an isomorphism

$$\Omega_{G/S} \to f^* e^* \Omega_{G/S}$$

as desired. By Morphisms, Lemma 0474 we have $\mathcal{C}_{S/G} \cong e^*\Omega_{G/S}$. If S is the spectrum of a field, then any $\mathcal{O}_S$-module on S is free and the final statement follows. $\square$